



# Modeling and Optimization of a Purchasing System in Uncertain Environments by an Integrated Fuzzy Business Process Simulation and Data Envelopment Analysis: A Novel Approach

A. Azadeh, S. Nassiri, M. Asadzadeh

Department of Industrial Engineering and Center of Excellence for Intelligent-Based Experimental Mechanics, College of Engineering University of Tehran, Tehran, Iran, P.O. Box 11155-4563

[azadeh@ut.ac.ir](mailto:azadeh@ut.ac.ir) [shimanassiri@ut.ac.ir](mailto:shimanassiri@ut.ac.ir) [smasadzadeh@ut.ac.ir](mailto:smasadzadeh@ut.ac.ir)

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## Abstract

In real situations particularly in business processes, the probability distributions are not known or imprecise and conventional modeling approaches may be questionable. Fuzzy simulation is capable of modeling uncertainties and vagueness. This paper introduces a novel approach based of fuzzy business process simulation (FBPS) and data envelopment analysis (DEA) for re-engineering, assessment and optimization of a purchasing management system. An actual purchasing system is studied and various data were collected through standard methods. The users of the purchasing management verified that the goals of a re-engineering model are to create a process-oriented approach in order to reduce purchasing process cycle time, faster customer response time and lower cost. The results of fuzzy simulation approach are compared with conventional simulation by t-test. It is shown the performance of the actual system is modeled and improved by the fuzzy simulation approach of this study. The results of fuzzy simulation are more accurate and similar to real situations than previous results by conventional simulation. In addition, DEA is used to identify optimal solution among proposed alternatives. This is the first study that introduces an integrated FBPS-DEA for optimization of purchasing management systems.

## 1. INTRODUCTION

Purchasing is increasingly being viewed as a key strategic function to help a firm to achieve competitive advantage. Usually data in purchasing and order processing tasks are poorly supported. The companies benefit from time and cost saving, increased flexibility, improved inter-organizational information sharing as well as the access to new markets. As indicated by [McIvor 1999] and [Macbet 1994] changes that will make a professional and advanced purchasing system include ensuring buyers as supply chain managers for supplying the final product to end customers.

Recent times have seen an interaction between the purchasing topic and other management objects by researchers and practitioners. Supply chain management has received increased attention during the past five to ten years, concurrently; the concept of strategic purchasing has also been emerging. For example based on [Carr & Smeltzer 1999] the results indicate that strategic purchasing is positively related to supplier responsiveness, changes in the supplier market, supplier communication and the firm's performance. Also in strategic management, the findings show purchasing departments shifting toward higher levels of strategic behavior (activities, concept of purchasing value-add, management style, training and development needs, concerns and purchasing strategy) [Cavinato 1999]. In other research, [Humphreys et al. 2000] apply Maister's Professional Service Firm model in an operational manner to outline the changing roles and responsibilities of the procurement function. Also based on [Wynstra & Weggeman 2003] it seems that three different levels (or areas) can be distinguished within purchasing integration in product development: (1) projects, (2) suppliers and (3) technologies.

In this study a new approach is introduced in order to optimize the purchasing system. Fuzzy approach, which is used widespread in

different context, is a suitable tool to handle ambiguousness and vagueness associated with real situations. Fuzzy simulation as a powerful approach was applied in various studies. [Buckley & Hayashi 1998] presented an application of fuzzy chaos to fuzzy simulation and applications of fuzzy simulation to fuzzy mathematical programming, fuzzy max flow, fuzzy regression, and tuning a fuzzy controller in their article. Also [Buckley 2005] in another article described fuzzy simulation and its various applications in modeling different systems. [Lui & Iwanura 2001] proposed a fuzzy simulation-based genetic algorithm for solving chance-constrained programming models with fuzzy decisions. As the specifications of a manufacturing system are often imperfectly known, [Gien & Jacquart 2005] designed a manufacturing system and used simulation techniques, based on fuzzy parameters, to verify the exactness of the design. [Maitia & Maiti 2006] proposed a multi-item inventory model with two-storage facilities was developed with advertisement, price and displayed inventory level-dependent demand in a fuzzy environment (purchase cost, investment amount and storehouse capacity are imprecise) and used fuzzy simulation as a part of their approach. [Puig et al. 2008] proposed a new approach for simulation of discrete linear time-invariant dynamic systems with fuzzy parameters and initial conditions. In their approaches they considered uncertain parameters as time-invariant during the simulation process.

Not only is the model presented in this study different from previous studies but it also is more efficient and accurate since DEA models have been used with respect to a set of comprehensive indicators to assess and optimize the purchasing system.

### 1.1. Business Process Simulation

Recent years have seen an increasing improvement in methodologies, techniques and tools to support the design of business processes under the banner of business process modeling (BPM). Although its origin can be traced back to the advent of scientific management around the early years of this century, it was only with the introduction of business process re-engineering (BPR) that BPM has emerged as a distinct field [Melao & Pidd 1999]. Business process simulation (BPS) is a dynamic approach for supporting analysis, improvement and re-engineering of business process via simulation modeling. [Melao & Pidd 1999] reviewed common issues for BPS and differences between the modeling approach of business processes and manufacturing processes. [Giaglis et al. 1999] studied the use of discrete simulation for business engineering. [Zarei 2001] presented the BPR simulation of a database management system and [Irani & Hlupic 2000] studied the re-engineering manufacturing processes through simulation modeling. A recent study presents an integrated simulation approach for modeling and assessment of information, business and production process of a manufacturer [Azadeh et al. 2008]. Furthermore, the hidden and concurrent effect of business, information and production processes are identified and improved.

### 1.2. Fuzzy Simulation

Recently fuzzy simulation is expanded for simulating different models, because in the real world we cannot obtain crisp data exactly and using fuzzy data as inputs for simulation results in more realistic outputs. A fuzzy age dependent replacement policy is considered in some of the following researches in which the lifetime of components is treated as fuzzy variables. To minimize the long-term expected cost

per unit time, a programming model is formulated. Also, a theorem for optimal solution existence is proposed. In order to solve the proposed model, a fuzzy simulation technique is designed which estimates the expected value of the objective function [Zhang et al. 2008]. A hybrid approach (here defined as ‘pseudo-fuzzy’) for the use of a classic discrete-event simulation for on-line production planning and control (thus in clearly transient Conditions) is presented to take several sources of uncertainty into account [Dassisti & Galantucci 2005]. Simulation techniques, based on fuzzy parameters, are used to verify the exactness of the design [Gien & Jacqmart 2005].

A fuzzy simulation-based genetic algorithm is provided for solving the proposed optimization problem [Huang 2008]. The use of simulation to reflect fuzzy inputs, which allows more complete probabilistic interpretation of model results, is demonstrated [Olson & Wu 2006].

The essential idea of fuzzy set is that to each element  $x$  of a universal set  $X$  is assigned a membership degree in  $[0, 1]$  of ownership to the set. Some of concepts which are used in fuzzy simulation approach are described below [Buckley 2004].

In this paper a fuzzy estimation from data using confidence intervals is considered. Let us describe the construction of our fuzzy estimators out of a set of confidence intervals. More detail can be found in [Jowers et al. 2007; Buckley 2004].

Alpha-cuts are slices through a fuzzy set producing regular (non-fuzzy) sets. If  $A$  is a fuzzy subset of some set  $X$ , then an  $\alpha$ -cut of  $A$ , written  $A[\alpha]$ , is defined as:

$$A[\alpha] = \{x \in X | A(x) \geq \alpha\} \text{ for all } \alpha, 0 < \alpha \leq 1$$

A value for  $\alpha$ -cut equal to 1 indicates certain knowledge whereas a value equal to 0 represents maximum uncertainty. In simulating systems, the parameters of probability distribution are assumed to be deterministic. In real life, this is not so. In fuzzy simulation methodology uncertainty is taken into account by replacing crisp parameters by fuzzy parameters. In fact in fuzzy simulation, it is assumed that the parameters are fuzzy numbers. We used center of area (COA) method for defuzzifying, because of its ease of use and popularity. Suppose that  $A$  is a (discrete) fuzzy subset of  $X$  and  $x \in A$  with the membership degree of  $A(x)$ , then the defuzzification COA method is defined as:

$$Defuzzy(A) = \frac{\sum_{x \in A} x \cdot A(x)}{\sum_{x \in A} A(x)}$$

Capital budgeting problem with fuzzy investment outlays, fuzzy annual net cash flows and fuzzy available investment capital, is studied based on credibility measure. According to this fact that the probability distribution parameters are associated with uncertainty and cannot be considered precisely, in this study we used fuzzy approach for simulating purchasing process. We choose to use a commercial discrete Event simulator and Response Surface Methodology to perform fuzzy simulation. In this paper we use fuzzy simulation for purchasing process management which is a novel and efficient approach and is not discussed in other related studies. The purchasing model will be explained in Section 3. The results of conventional and fuzzy simulation are shown in Section 5. In Section 5.1 and 5.2 it is proved that the fuzzy simulation could model the behavior of mentioned system more precise than the conventional simulation proposed in [Azadeh & Harandi 2009] and it optimizes the purchasing system according to the DEA model presented, therefore its results are more reliable.

## 2. DATA ENVELOPMENT ANALYSIS (DEA)

Consistent with DEA terminology, the term decision making unit (DMU) refers to the individuals in the evaluation group. DEA generates a surface called the frontier that follows the peak performers and envelops the remainder [Charnes et al. 1978].

The four basic DEA models are the Charnes, Cooper and Rhodes (CCR) model, the Banker, Charnes and Cooper (BCC) model, the multiplicative model, and the additive model. Only the CCR will be discussed here. These models can be distinguished by the envelopment surface and the orientation [Banker et al. 1984].

The objective of DEA is to obtain the weights that maximize the efficiency of the DMU under evaluation, while limiting the efficiency of all DMUs to less than or equal to 1.0 (for input-oriented models). [Charnes et al. 1978] recognized the difficulty in seeking a standard set of weights to calculate the relative efficiency, which might cause DMUs to value inputs and outputs differently and thus adopt different weights. Therefore, they proposed that each DMU should adopt weights which show it in the most favorable light relative to the other DMUs.

### 2.1. Basic Models of DEA

The original fractional CCR model (1) evaluates the relative efficiencies of  $n$  DMUs ( $j=1, \dots, n$ ), each with  $m$  inputs and  $s$  outputs denoted by  $x_{1j}, x_{2j}, \dots, x_{mj}$  and  $y_{1j}, y_{2j}, \dots, y_{sj}$  respectively. This is done so by maximizing the ratio of weighted sum of output to the weighted sum of inputs:

$$\begin{aligned} \text{Max } \theta &= \frac{\sum_{r=1}^s u_r y_{r0}}{\sum_{i=1}^m v_i x_{i0}} \\ \text{s.t. } \frac{\sum_{r=1}^s u_r y_{rj}}{\sum_{i=1}^m v_i x_{ij}} &\leq 1 \quad j=1, \dots, n, \quad r=1, \dots, s \quad (1) \\ u_r, v_i &\geq 0 \quad i=1, \dots, m, \quad r=1, \dots, s \end{aligned}$$

In model (1), the efficiency of DMU<sub>0</sub> is  $\theta_0$ , and  $u_r$  and  $v_i$  are the factor weights. In this model, the relative efficiencies of DMUs are maximized by maximizing the ratio of weighted sum of outputs to the weighted sum of inputs. It should be noted that the mentioned ratio cannot exceed 1, because this ratio expresses the efficiency of DMUs. However, for computational convenience the fractional programming model (1) is re-expressed in linear programming (LP) form as follows:

$$\begin{aligned}
\text{Max } \theta &= \sum_{r=1}^s u_r y_{r0} \\
\text{s.t. } &\sum_{r=1}^s u_r y_{rj} - \sum_{i=1}^m v_i x_{ij} \leq 0 \quad j = 1, \dots, n \\
&\sum_{i=1}^m v_i x_{i0} = 1 \quad (2) \\
&u_r, v_i \geq \varepsilon \quad i = 1, \dots, m, \quad r = 1, \dots, s
\end{aligned}$$

Where  $\varepsilon$  is a non-Archimedean infinitesimal introduced to ensure that all the factor weights will have positive values in the solution. The model (3) evaluates the relative efficiencies of  $n$  DMUs ( $j = 1 \dots n$ ) respectively, by minimizing inputs when outputs are constant. The dual of linear program (LP) model for input oriented (which is used in this study) CCR is as follows [Banker et al. 1986]:

$$\begin{aligned}
\text{Min } \theta \\
\text{s.t. } &\theta x_{i0} \geq \sum_{j=1}^n \lambda_j x_{ij} \quad i = 1, \dots, m \\
&y_{r0} \leq \sum_{j=1}^n \lambda_j y_{rj} \quad r = 1, \dots, s \quad (3) \\
&\lambda_j \geq 0
\end{aligned}$$

If  $\sum \lambda_j = 1$  ( $j=1 \dots n$ ) is added to model (3), the BCC model is obtained which is input oriented and its return to scale is variable [Azadeh et al. 2003]. The calculations provide a maximal performance measure using piecewise linear optimization on each DMU with respect to the closest observation on the frontier.

The efficiency of each branch will be defined and according to Jahanshahloo model (4) the ranking of bank branches will be computed.

$$\begin{aligned}
\text{Min } Z_0 &= \omega_0 - \varepsilon \left( \sum_{i=1}^2 s_i^- + \sum_{r=1}^8 s_r^+ \right) \\
\text{s.t. } &\sum_{j=1}^n \lambda_j x_{ij} + s_i^- = x_{i0} + \omega_0 \\
&j \neq 0 \\
&\sum_{j=1}^n \lambda_j y_{rj} + s_r^+ = y_{r0} - \omega_0 \\
&j \neq 0 \\
&\sum_{j=1}^n \lambda_j = 1 \\
&j \neq 0 \\
&\lambda_j, s_i^-, s_r^+ \geq 0 \quad \forall j (j \neq 0) \text{ and } \omega_0 \text{ is free}
\end{aligned} \quad (4)$$

### 3. METHOD: THE PURCHASING SYSTEM

The integrated model is presented in Figure 1. First, data is collected and analyzed through robust statistical (parametric) and fuzzy statistics. Then, conceptual models of the purchasing system are developed by integrated definition (IDEF0) or similar approaches. Conventional and fuzzy simulation models are developed by incorporating related parameters and variables. The simulation models are then verified and validated through t-tests. Also, t-test is used to foresee if fuzzy simulation approach provides better estimation than conventional simulation. Next, the outputs of fuzzy and conventional simulations are inputted to DEA models for ranking and optimization. Finally, the best configuration is evaluated through DEA.

#### 3.1. Data Collection and Analysis

Data related to processes and activities were collected and analyzed by Kolmogorov-Smirnov goodness of fit test [Law & Kelton 1991]. The best distribution function was identified with lowest error square and highest P-value [Triola 2000]. A p-value of 0.05 is used for rejection criterion. Table 1 presents the most fitted distribution function for six major activities of the purchasing system. For other process information, interview and questionnaire techniques were used.

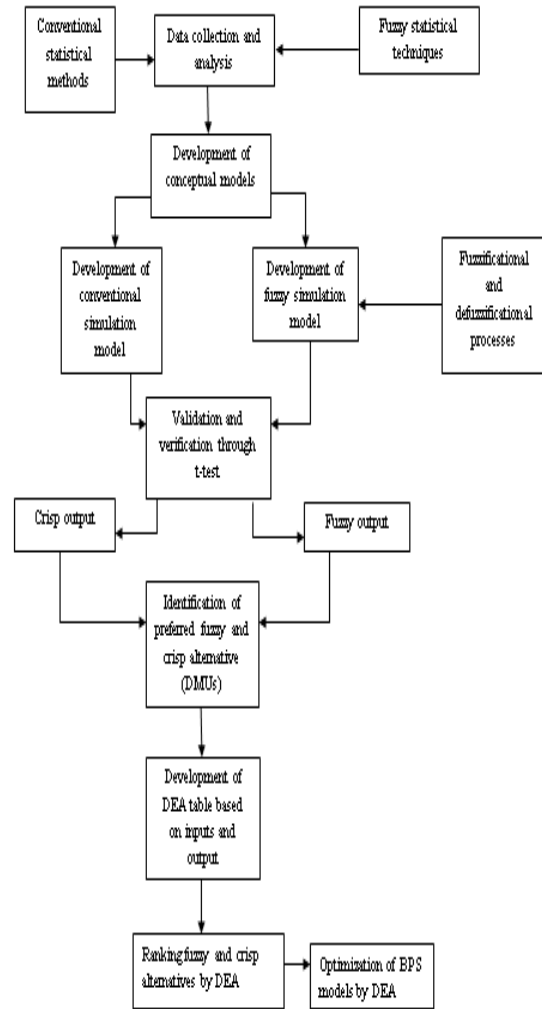


Figure 1. The proposed integrated fuzzy model

**Table 1.** The selected distributions of major activities

| Tested distribution<br>functions<br>Activities | Choice the lowest error<br>square |                                   |
|------------------------------------------------|-----------------------------------|-----------------------------------|
|                                                | P-<br>Value                       | Selected<br>distribution function |
| Time between two-monthly orders (days)         | > 0.15                            | 0.1 + LOGN (0.159 , 0.0844 )      |
| Time between direct orders (days)              | > 0.15                            | 0.03 + LOGN (0.816 , 0.598 )      |
| Cost of two-monthly orders (\$)                | > 0.15                            | LOGN (3.28,2.29)                  |
| Cost of direct orders (\$)                     | > 0.15                            | 1<br>ERLAN(25.3 , 1) +            |
| Total cost of cash expense account (\$)        | > 0.15                            | NORM (109 , 68)                   |
| Number of credit expense account               | > 0.15                            | 0.5<br>WEIB(3.21 , 0.954) +       |

#### 4. EXPERIMENT: THE FBPS-DEA

Simulation models provide quantitative information that can be used for decision-making and can be regarded as problem understanding rather than problem solving tools. According to [Pidd 1998] BPS can be viewed as three phases: (1) model building, (2) computer programming and (3) experimentation on the model.

The business process simulation model was developed by Visual SLAM developed by [Pritsker et al. 1997]. The purchase orders (direct and two-monthly) are considered as entities and time between creation of orders and cost of orders are defined as variables in the simulation model. Resources and queues are defined as buyers, inventory controllers and inspectors and employees in purchasing, financial, planning and operational departments. Time and human resource requirements in different stages of the process were obtained by conventional work-study and are introduced as model parameters.

A number of random variables and an experimental design were employed to gain 95% confidence interval for all measures of performance including average time of purchasing process cycle for direct and two-monthly orders, inventory level and human resources under steady-state conditions. Furthermore, 40 replication of the model was run to provide high level of confidence. Each run included 250 days of which 44 days were eliminated due to prior identification of transient state (warm-up period).

##### 4.1. Validation and Verification

Conceptual model validation was performed to ensure that the assumptions underlying the conceptual model are correct and that the representation of the problem is reasonable for the stated goals of the study. First, the IDEF0 and flow chart models were logically validated and approved by the process engineers. Subsequently, validation and verification were parts of model development process, using a combination of subjective and objective methods. Subjective tests involve people who are knowledgeable about one or more aspects of the system, making judgment about the model and its behavior. Here different techniques were used, such as tracing orders (entities) through the departments in the simulation using Trace and MONTR control and checking variables in simulation different times to see the program was operating as intended. Objective tests require comparison of data on the system's behavior and corresponding data generated by the BPS

model. Therefore, historical input-output data for the last two years operation of the purchasing process was used to test the model. Moreover, time between settling cash expense accounts and cycle time of purchasing process for two-monthly and direct orders in current system and simulation model were compared by t-tests and F-test and consequently the simulation model was validated. More details of the analysis can be found in [Azadeh et al. 2006].

##### 4.2. Fuzzy Calculation

Consider X a random variable with probability density function  $N(\mu, \sigma^2)$ , with unknown mean  $\mu$  and known variance  $\sigma^2$ . For unknown variance see [Buckley 2004]. To estimate  $\mu$  a random sample  $X_1, \dots, X_n$  from  $N(\mu, \sigma^2)$  is obtained. Suppose the mean of this random sample turns out to be  $\bar{X}$ , which is a crisp number, not a fuzzy number. It is

known that  $\bar{X}$  is  $N(\mu, \frac{\sigma^2}{n})$  so  $\frac{\bar{X} - \mu}{\sigma/\sqrt{n}}$  is  $N(0, 1)$ . Therefore P (-

$$Z_{\beta/2} \leq \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \leq Z_{\beta/2} = 1 - \beta, \text{ where } Z_{\beta/2} \text{ is the Z value so that}$$

the probability of an  $N(0, 1)$  random variable exceeding it is  $\beta/2$ .

$$\text{Now solve the inequality for } \mu \text{ producing } P(\bar{X} - Z_{\beta/2} \frac{\sigma}{\sqrt{n}} \leq \mu \leq \bar{X} + Z_{\beta/2} \frac{\sigma}{\sqrt{n}}) = 1 - \beta.$$

This lead directly to the  $(1 - \beta)$  100% confidence interval for  $\mu$

$$[\theta_1(\alpha), \theta_2(\alpha)] = (\bar{X} - Z_{\beta/2} \frac{\sigma}{\sqrt{n}}, \bar{X} + Z_{\beta/2} \frac{\sigma}{\sqrt{n}}), \text{ where}$$

$Z_{\beta/2}$  is defined as

$$Z_{\beta/2} = \int_{-\infty}^{\infty} N(0,1) dx = 1 - \frac{\beta}{2}$$

$N(0, 1)$  denotes the normal density with mean zero and unit variance. Put these confidence intervals together as discussed above and we obtain  $\tilde{\mu}$  our fuzzy estimator of  $\mu$ . Consider X a random variable with probability density function  $f(x; \theta)$ .

If  $n \geq 30$  normal distribution should be used, otherwise we use t-student distribution. Namely the  $(1 - \beta)$  100% confidence interval for  $\mu$

$$\text{is } [\theta_1(\alpha), \theta_2(\alpha)] = (\bar{X} - t_{\beta/2} \frac{s}{\sqrt{n}}, \bar{X} + t_{\beta/2} \frac{s}{\sqrt{n}}). \text{ The}$$

intervals for variables are shown in Table 2.

**Table 2.** The confidence intervals for different  $\alpha$ -cuts

| Variables                               | Confidence interval for $\alpha = 0.01$ | Confidence interval for $\alpha = 0.2$ | Confidence interval for $\alpha = 0.5$ | Confidence interval for $\alpha = 0.8$ |
|-----------------------------------------|-----------------------------------------|----------------------------------------|----------------------------------------|----------------------------------------|
| Time between two-monthly orders (days)  | (0.0896,0.22)                           | (0.098,0.205)                          | (0.158,0.199)                          | (0.161,0.196)                          |
| Time between direct orders (days)       | (0.294,1.338)                           | (0.412,1.21)                           | (0.455,1.167)                          | (0.483,1.139)                          |
| Cost of two-monthly orders (\$)         | (46.044, 689.809)                       | (206.985, 528.868)                     | (283.089, 452.764)                     | (335.489, 400.364)                     |
| Cost of direct orders (\$)              | (12.972, 39.597)                        | (19.628, 32.94)                        | (22.776, 29.793)                       | (24.943, 27.626)                       |
| Total cost of cash expense account (\$) | (57.014, 162.479)                       | (85.922, 133.571)                      | (97.489, 122.004)                      | (105.176, 114.316)                     |
| Number of credit expense account        | (0.976, 6.595)                          | (2.526, 5.044)                         | (3.138, 4.433)                         | (3.544, 4.027)                         |

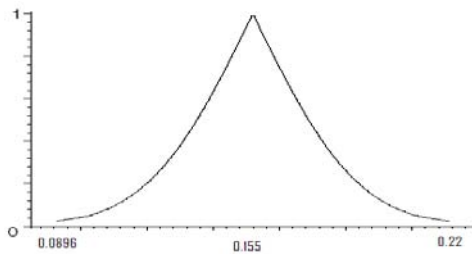
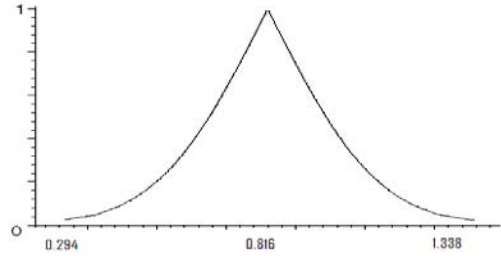
## 5. RESULTS AND ANALYSIS

### 5.1. Fuzzy Simulation of Purchasing Process

In, general when a time-dependent distribution is considered, fuzzy simulation is useful and lead to better results. Thus, in Time between two-monthly orders and Time between direct orders, distributions  $\mu_1$  and  $\mu_2$  are considered as a fuzzy number (Figures 2 and 3). To fuzzify  $\mu_1$  and  $\mu_2$  the method mentioned above is used, and the following values are obtained as follows:

$$\tilde{\mu}_1 = (0.0896, 0.155, 0.22) ,$$

$$\tilde{\mu}_2 = (0.294, 0.816, 1.338)$$

**Figure 2.** Shape of fuzzy number  $\tilde{\mu}_1$ **Figure 3.** Shape of fuzzy number  $\tilde{\mu}_2$ 

Then, to obtain the values of output as a fuzzy number we must run the simulation for lower bounds, mean values, upper bounds of the inputs, it means the simulation model should be run 3 times, Thus, 3 values for outputs are resulted, after that outputs are defuzzified with COA method. Hence, one value for the outputs is obtained.

The results of 18 outputs from fuzzy simulated model for the “time between settling cash expense accounts” which were obtained from a 200 days period randomly and the data of real-world which were obtained from “ cycle time of purchasing process for two-monthly and direct orders in current system”, are shown in Table 3.

Now, to validate the fuzzy model, the hypothesis of

**Table 3.** Time between settling cash expense accounts

|                             |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
|-----------------------------|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Fuz<br>zy<br>sim<br>ulation | 2 | 4 | 2 | 5 | 4 | 1 | 2 | 6 | 9 | 2 | 2 | 2 | 3 | 4 | 3 | 5 | 2 | 4 |
| Act<br>ual<br>data          | 3 | 6 | 2 | 4 | 4 | 1 | 0 | 8 | 1 | 1 | 2 | 3 | 4 | 6 | 1 | 6 | 3 | 4 |

equivalence variances for fuzzy outputs and real data is accepted at the level of 95% confidence and the hypothesis of equivalence means is also accepted at the level of 95% confidence. The hypothesis was tested using SPSS software and the results are shown in Table 4 and 5. To make the data applicable for T-test the extreme data are crossed out.

**Table 4.** Independent sample test for fuzzy simulation versus actual data

| Alternative                              | N  | Mean   | Standard Deviation | Standard Error of Mean |
|------------------------------------------|----|--------|--------------------|------------------------|
| Actual System                            | 17 | 3.4118 | 2.18114            | .52900                 |
| Fuzzy Business Process Simulation (FBPS) | 17 | 3.1176 | 1.40900            | .34173                 |

**Table 5.** The results of t-test for hypothesis testing of means (actual versus fuzzy simulation)

| F   | Significance | t-value | Degrees of Freedom | Significance (2-tailed) | Mean Difference | Standard Error of Difference | 95% Confidence Interval of the Difference |     |
|-----|--------------|---------|--------------------|-------------------------|-----------------|------------------------------|-------------------------------------------|-----|
| 2.3 | .134         | 0.46    | 32                 | 0.64                    | 0.29            | 0.629                        | -0.988                                    | 1.5 |

Finally, to compare fuzzy simulation and crisp simulation as is shown in Table 6 (outputs of fuzzy and crisp simulation) the following is considered:

$D_{1i}$  = [real world data-crisp simulation outputs]  $i = 1, 2, \dots, 18$

$D_{2i}$  = [real world data- Fuzzy simulation outputs]  $i = 1, 2, \dots, 18$

First, the equivalence of variances is tested. Then the following hypothesis tested for means as follows:

$$\begin{cases} H_0 : \mu_{D2} \geq \mu_{D1} \\ H_1 : \mu_{D2} < \mu_{D1} \end{cases}$$

**Table 6.** Time between settling cash expense accounts for fuzzy versus conventional (crisp) simulation

|                         |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
|-------------------------|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| <b>Fuzzy simulation</b> | 2 | 4 | 2 | 5 | 4 | 1 | 2 | 6 | 9 | 2 | 2 | 2 | 3 | 4 | 3 | 5 | 2 | 4 |
| <b>Crisp simulation</b> | 1 | 3 | 2 | 5 | 4 | 1 | 3 | 5 | 4 | 3 | 2 | 1 | 0 | 2 | 6 | 6 | 3 | 6 |

Hypothesis of the equivalence of variances means and  $H_0$  (in hypothesis testing) are rejected. This means that the fuzzy simulation output is smaller than that of conventional simulation. Moreover, the fuzzy simulation outputs are closer to real-world rather than crisp simulation. The results are shown in Tables 7 and 8.

**Table 7.** Independent Samples Test for fuzzy simulation versus conventional simulation

| Group                   | N  | Mean   | Standard Deviation | Standard Error of Mean |
|-------------------------|----|--------|--------------------|------------------------|
| <b>Fuzzy simulation</b> | 18 | 2.0556 | 1.89340            | .44628                 |
| <b>Crisp simulation</b> | 18 | 1.0000 | .76696             | .18078                 |

**Table 8.** Output for testing variances equivalence and testing on DEA method. best results and hypothesis

| F    | Sig. | t    | Df   | Sig. (2-tailed) | Mean Difference | Std Error Difference | 95% Confidence Interval of the Difference |       |
|------|------|------|------|-----------------|-----------------|----------------------|-------------------------------------------|-------|
|      |      |      |      |                 |                 |                      | Lower                                     | Upper |
| 11.5 | .002 | 2.19 | 34   | .035            | 1.055           | .481                 | .077                                      | 2.03  |
|      |      | 2.19 | 22.4 | .039            | 1.055           | .481                 | .058                                      | 2.05  |

## 5.2. DEA Results

In our applied DEA model we have 20 DMUs where,  $DMU_1$  Indicates the fuzzy approach for purchasing system and  $DMU_2$  presents the conventional simulation approach. The other DMUs show the different linear combinations of these DMUs since in real environments it is probable to face uncertainty in part of a system. The inputs and output of these DMUs (3-20) are calculated in equations (5)

and (6) respectively where  $\alpha_j$  is the proportion of fuzziness in the cited combination which puts on the values between 0.05 and 0.95 increased by 0.05.

$$X_{ij} = \alpha_j X_{i1} + (1 - \alpha_j) X_{i2} \quad j = 3, \dots, 20 \quad (5)$$

$$Y_{ij} = \alpha_j Y_{i1} + (1 - \alpha_j) Y_{i2} \quad j = 3, \dots, 20 \quad (6)$$

Table 9 shows raw input and output data for DEA. The first DMU indicates the fuzzy approach and the data have been derived from the estimated distributions in Table 2. The defuzzified data is used as raw data for DEA. Table 10 presents the technical efficiency in the first column based on input oriented CCR model for each branch, and the surplus and slacks in the remaining columns. The efficiency score is between zero and one. Hence those branches with efficiency of one are efficient. The week efficient DMUs have non-zero slacks, surplus and the optimum objective value of one while the strong efficient DMUs have zero slacks, surplus and the optimum objective value of one.

According to model 3 results, an optimization procedure is presented. This procedure utilizes the values of slacks, surpluses and efficiency scores in optimal solution of Model 3. Let us define a virtual DMU as a target for each of which. The inputs and output of this virtual target DMU are obtained from equation (7).

$$\begin{aligned} \hat{x}_{i0} &= \theta x_{i0} - s_i & i &= 1, \dots, m \\ \hat{y}_{r0} &= y_{r0} + s_r & r &= 1, \dots, s \end{aligned} \quad (7)$$

Where  $\hat{x}$  and  $\hat{y}$  are target values,  $s_i$  and  $s_r$  are slack and surpluses in Model (3) and  $\theta$  is optimum objective value (efficiency score) of Model (3). From efficiency improvement point of view the target values of input and output in equation (7) are the optimal values and an optimization procedure can be described as follows:

An inefficient DMU should reduce its inputs to target inputs and increase its output to target output. A week efficient DMU should reduce or increase its inputs and outputs respectively by slacks and surpluses. The results in Table 10 indicate the values of slacks and surpluses.

Table 11 presents the ranking results of the branches based As is shown in these tables the fuzzy approach ( $DMU_1$ ) obtained optimized purchasing system. Figure 4 also depicts the superiority of fuzziness in our BPS model according to the efficiency scores.

**Table 9.** The raw data for input and output indicators

| Input/<br>Output<br><br>Model | Inputs                                 |                                   |                                 |                            |                                         |                                  | Output                                             |
|-------------------------------|----------------------------------------|-----------------------------------|---------------------------------|----------------------------|-----------------------------------------|----------------------------------|----------------------------------------------------|
|                               | Time between two-monthly orders (days) | Time between direct orders (days) | Cost of two-monthly orders (\$) | Cost of direct orders (\$) | Total cost of cash expense account (\$) | Number of credit expense account | Time between settling cash expense accounts (days) |
| DMU <sub>1</sub>              | 0.15                                   | 0.81                              | 367.92                          | 26.28                      | 109.74                                  | 3.78                             | 3.44                                               |
| DMU <sub>2</sub>              | 1.27                                   | 2.73                              | 372                             | 26.3                       | 109                                     | 3.77                             | 3.16                                               |
| DMU <sub>3</sub>              | 0.71                                   | 1.77                              | 369.96                          | 26.29                      | 109.3                                   | 3.77                             | 3.3                                                |
| DMU <sub>4</sub>              | 0.654                                  | 1.674                             | 369.756                         | 26.289                     | 109.407                                 | 3.7755                           | 3.314                                              |
| DMU <sub>5</sub>              | 0.6                                    | 1.58                              | 369.6                           | 26.3                       | 109.44                                  | 3.78                             | 3.328                                              |
| DMU <sub>6</sub>              | 0.542                                  | 1.482                             | 369.348                         | 26.287                     | 109.481                                 | 3.7765                           | 3.342                                              |
| DMU <sub>7</sub>              | 0.486                                  | 1.386                             | 369.144                         | 26.286                     | 109.518                                 | 3.777                            | 3.356                                              |
| DMU <sub>8</sub>              | 0.43                                   | 1.29                              | 368.94                          | 26.285                     | 109.555                                 | 3.7775                           | 3.37                                               |
| DMU <sub>9</sub>              | 0.374                                  | 1.194                             | 368.736                         | 26.284                     | 109.592                                 | 3.778                            | 3.384                                              |
| DMU <sub>10</sub>             | 0.318                                  | 1.098                             | 368.532                         | 26.283                     | 109.629                                 | 3.7785                           | 3.398                                              |
| DMU <sub>11</sub>             | 0.262                                  | 1.002                             | 368.328                         | 26.282                     | 109.666                                 | 3.779                            | 3.412                                              |
| DMU <sub>12</sub>             | 0.206                                  | 0.906                             | 368.124                         | 26.281                     | 109.703                                 | 3.7795                           | 3.426                                              |
| DMU <sub>13</sub>             | 0.822                                  | 1.962                             | 370.368                         | 26.292                     | 109.296                                 | 3.774                            | 3.272                                              |
| DMU <sub>14</sub>             | 0.766                                  | 1.866                             | 370.164                         | 26.291                     | 109.333                                 | 3.7745                           | 3.286                                              |
| DMU <sub>15</sub>             | 0.934                                  | 2.154                             | 370.776                         | 26.294                     | 109.222                                 | 3.773                            | 3.244                                              |
| DMU <sub>16</sub>             | 0.878                                  | 2.058                             | 370.572                         | 26.293                     | 109.259                                 | 3.7735                           | 3.258                                              |
| DMU <sub>17</sub>             | 1.046                                  | 2.346                             | 371.184                         | 26.296                     | 109.148                                 | 3.772                            | 3.216                                              |
| DMU <sub>18</sub>             | 0.99                                   | 2.25                              | 370.98                          | 26.295                     | 109.185                                 | 3.7725                           | 3.23                                               |
| DMU <sub>19</sub>             | 1.158                                  | 2.538                             | 371.592                         | 26.298                     | 109.074                                 | 3.771                            | 3.188                                              |
| DMU <sub>20</sub>             | 1.102                                  | 2.442                             | 371.388                         | 26.297                     | 109.111                                 | 3.7715                           | 3.202                                              |

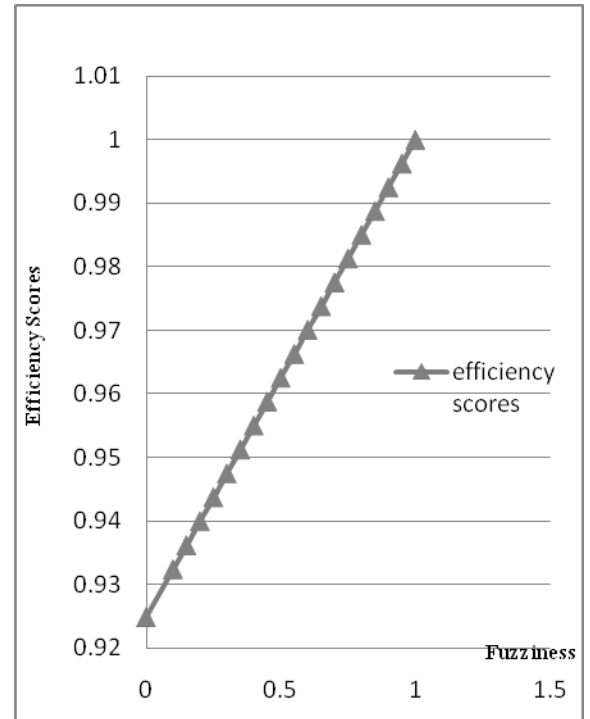
**Table 10.** Surplus and slacks of model (3)

| DMU name          | Efficiency Score | Input1   | Input2   | Input3   | Input4   | Input5 | Input6   | Output |
|-------------------|------------------|----------|----------|----------|----------|--------|----------|--------|
| DMU <sub>1</sub>  | 1                | 0        | 0        | 0        | 0        | 0      | 0        | 0      |
| DMU <sub>10</sub> | 0.988790841      | 0.166267 | 0.285582 | 0.973113 | 0.02925  | 0      | 0.002297 | 0      |
| DMU <sub>11</sub> | 0.992529749      | 0.111264 | 0.191108 | 0.651195 | 0.019574 | 0      | 0.001537 | 0      |
| DMU <sub>12</sub> | 0.996266134      | 0.055841 | 0.095914 | 0.326823 | 0.009824 | 0      | 0.000772 | 0      |
| DMU <sub>13</sub> | 0.955026759      | 0.642358 | 1.103321 | 3.759537 | 0.113005 | 0      | 0.008876 | 0      |
| DMU <sub>14</sub> | 0.95878848       | 0.591147 | 1.015361 | 3.459816 | 0.103996 | 0      | 0.008168 | 0      |
| DMU <sub>15</sub> | 0.94749567       | 0.743507 | 1.277057 | 4.351538 | 0.1308   | 0      | 0.010273 | 0      |
| DMU <sub>16</sub> | 0.95126249       | 0.693145 | 1.190553 | 4.056778 | 0.12194  | 0      | 0.009577 | 0      |
| DMU <sub>17</sub> | 0.93995437       | 0.84296  | 1.447877 | 4.933604 | 0.148296 | 0      | 0.011647 | 0      |
| DMU <sub>18</sub> | 0.943726298      | 0.793446 | 1.362832 | 4.643814 | 0.139585 | 0      | 0.010963 | 0      |
| DMU <sub>19</sub> | 0.932402836      | 0.940711 | 1.615776 | 5.505714 | 0.165493 | 0      | 0.012998 | 0      |
| DMU <sub>2</sub>  | 0.92484105       | 1.036757 | 1.780746 | 6.067847 | 0.182389 | 0      | 0.014325 | 0      |
| DMU <sub>20</sub> | 0.936179883      | 0.892048 | 1.532192 | 5.220905 | 0.156932 | 0      | 0.012326 | 0      |
| DMU <sub>3</sub>  | 0.962547657      | 0.539513 | 0.926674 | 3.157619 | 0.094913 | 0      | 0.002642 | 0      |
| DMU <sub>4</sub>  | 0.96630429       | 0.487457 | 0.837262 | 2.852949 | 0.085755 | 0      | 0.006735 | 0      |
| DMU <sub>5</sub>  | 0.970093839      | 0.43694  | 0.74912  | 2.605474 | 0.089096 | 0      | 0.010024 | 0      |
| DMU <sub>6</sub>  | 0.97380994       | 0.382078 | 0.656262 | 2.236196 | 0.067216 | 0      | 0.005279 | 0      |
| DMU <sub>7</sub>  | 0.977558961      | 0.328756 | 0.564676 | 1.924118 | 0.057836 | 0      | 0.004543 | 0      |
| DMU <sub>8</sub>  | 0.98130545       | 0.275014 | 0.472367 | 1.609577 | 0.048381 | 0      | 0.0038   | 0      |
| DMU <sub>9</sub>  | 0.985049409      | 0.22085  | 0.379335 | 1.292574 | 0.038853 | 0      | 0.003052 | 0      |

## 6. CONCLUSIONS

Previous studies mostly consider simulation modeling of production system in various settings. Moreover, recently simulation

of business systems have been discussed and highlighted by several researchers. Simulations of information and business systems have been considered in context of business process simulation (BPS),

**Figure 4.** The correlation between efficiency scores and fuzziness**Table 11.** Ranking results of the units based on DEA method

| DMU               | Efficiency  | Rank |
|-------------------|-------------|------|
| DMU <sub>1</sub>  | 1.038488752 | 1    |
| DMU <sub>10</sub> | 0.99440109  | 4    |
| DMU <sub>11</sub> | 0.996267393 | 3    |
| DMU <sub>12</sub> | 0.998133697 | 2    |
| DMU <sub>13</sub> | 0.977604359 | 13   |
| DMU <sub>14</sub> | 0.979470662 | 12   |
| DMU <sub>15</sub> | 0.973871752 | 15   |
| DMU <sub>16</sub> | 0.975738055 | 14   |
| DMU <sub>17</sub> | 0.970139145 | 17   |
| DMU <sub>18</sub> | 0.972005448 | 16   |
| DMU <sub>19</sub> | 0.966406538 | 19   |
| DMU <sub>2</sub>  | 0.962673931 | 20   |
| DMU <sub>20</sub> | 0.968272842 | 18   |
| DMU <sub>3</sub>  | 0.981336966 | 11   |
| DMU <sub>4</sub>  | 0.983203269 | 10   |
| DMU <sub>5</sub>  | 0.985087797 | 9    |
| DMU <sub>6</sub>  | 0.986935876 | 8    |
| DMU <sub>7</sub>  | 0.988802179 | 7    |
| DMU <sub>8</sub>  | 0.990668483 | 6    |
| DMU <sub>9</sub>  | 0.992534786 | 5    |

computerized network simulation (CNS) and conventional information simulation.

In this study a purchasing process in a research-manufacturing industry was modeled and analyzed by an integrated fuzzy business process simulation and data envelopment analysis. BPS was used as an experimental vehicle for better understanding of the purchasing system, and then fuzzy logic was used for simulating model of purchasing process, which was prepared by Visual Slam. A comparison between crisp simulation and fuzzy simulation was performed by t-test. Results showed that outputs of fuzzy simulation

are closer to the actual data than crisp simulation. It was concluded that fuzzy modeling provides more accurate estimation for the purchasing systems with data and environmental ambiguity. In addition, an integrated model of data envelopment analysis (DEA) for finding optimal solution among proposed alternatives was presented. The DEA results indicated the superiority of fuzzy simulation comparing to other alternatives.

The proposed new approach of this study for purchasing systems with data ambiguity is compared with the recent studies in this area and it is superior to the stated studies in several features.

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